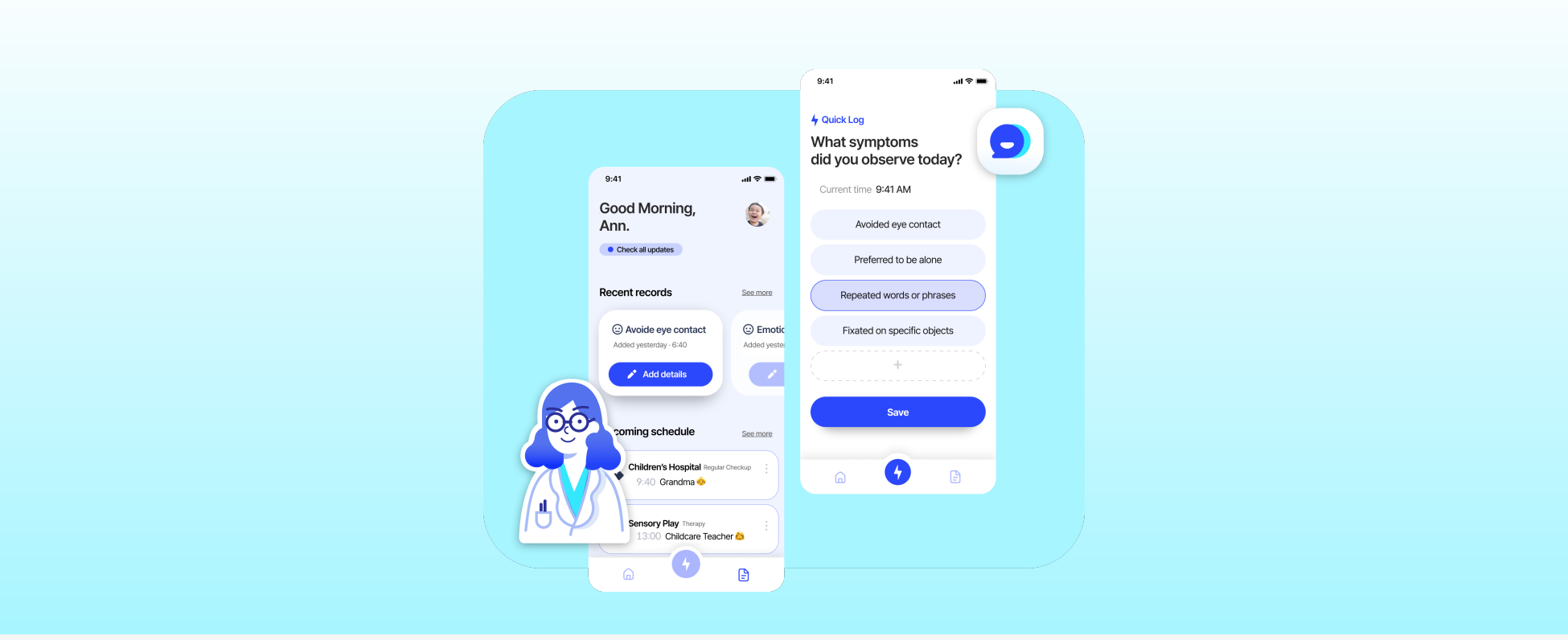


HEALTHCARE · 2019 - 2021

KongDoc - Diary App for Children with ASD

How can children with ASD recieve accurate and continuous care over time?

MY ROLE	THE TEAM
UX Designer & Project Manager	2 Medical Doctors, 2 Software Engineers

AWARDS

2019 - Dreamshare Medical Hackathon
2020 - Yonsei Start-up Incubation Program
2020 - Digital Health Hackathon (1st Place)
2021 - Kakao Impact Incubation Program

OVERVIEW

I worked on interacton design of Ballie, building trust and emotional connections with users.

At the Healthcare Entrepreneurship Society "Medilux," multi-background members interested in children with developmental disabilities gathered. Through item development, our team won the grand prize at the Digital Health Healthcare Hackathon and received project development investment.

- Recognizable Greetings
- Automatic Charging
- Adaptable Screen Projections
- Multi-Modal Experiences

BACKGROUND

My interest began through working with children with ASD.

Through volunteering with diverse groups of children, I developed a deep interest in early childhood development. By working with both typically developing children and children with autism spectrum disorder (ASD), I became increasingly aware of the challenges their parents face, particularly in accessing timely and effective support.

High Treatment Costs & Long Wait Times

The cost of effective autism treatment is extremely high. Applied Behavior Analysis (ABA), one of the most widely recognized treatment methods, can cost up to ₩4 million per month for full-time sessions. (Autism Partnership Korea)

Rising Prevalence of ASD

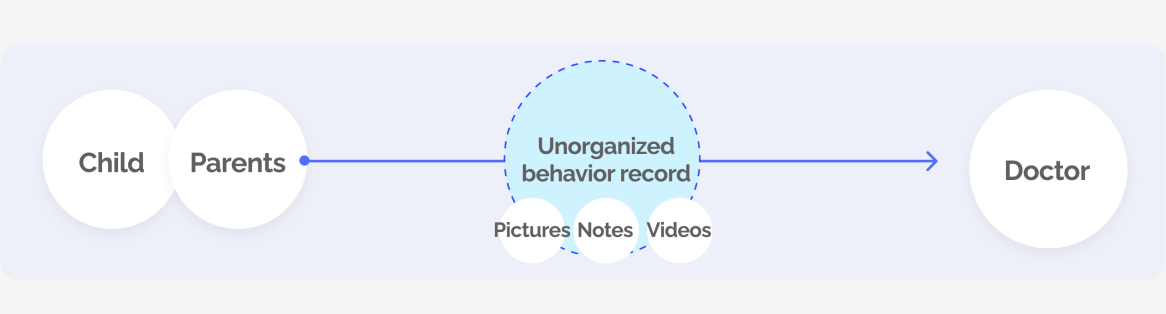
According to the CDC, the prevalence of autism spectrum disorder in the United States has risen sharply, from 1 in 166 children (0.67%) in 2004 to 1 in 68 (1.46%) in 2016, and further to 1 in 59 (1.69%) in 2018.

Importance of Early Intervention

According to the CDC, the prevalence of autism spectrum disorder in the United States has risen sharply, from 1 in 166 children (0.67%) in 2004 to 1 in 68 (1.46%) in 2016, and further to 1 in 59 (1.69%) in 2018.

PROBLEM

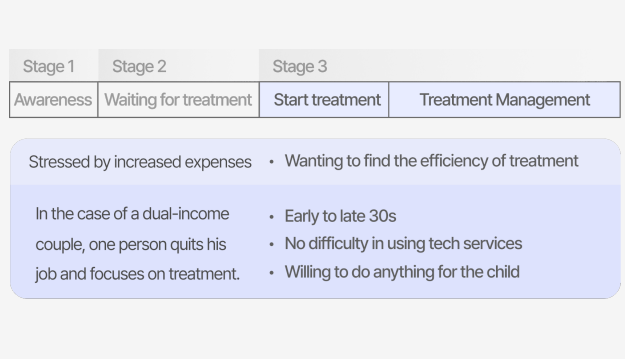
Fragmented behavior records disrupt clinical interpretation and continuity of care.


Child
Parents
Unorganized behavior record
Pictures
Notes
Videos
Doctor

Among the various issues related to developmental disabilities, we focused on behavioral records of children with developmental disabilities. For children with developmental disabilities, everyday behavioral observations often play an important role in clinical assessment and ongoing care. Caregivers often document behaviors through scattered photos, videos, and notes, making it difficult to identify patterns and communicate them efficiently during clinical visits.

TARGET USER GROUP

Caregivers during the initiation and management phase of developmental therapy



Stage 1	Stage 2	Stage 3	
Awareness	Waiting for treatment	Start treatment	Treatment Management

Stressed by increased expenses
- Wanting to find the efficiency of treatment

In the case of a dual-income couple, one person quits his job and focuses on treatment.
- Early to late 30s
- No difficulty in using tech services
- Willing to do anything for the child

Caregivers' needs and decision-making priorities shift depending on their psychological state, parenting stage, and position within the treatment journey.

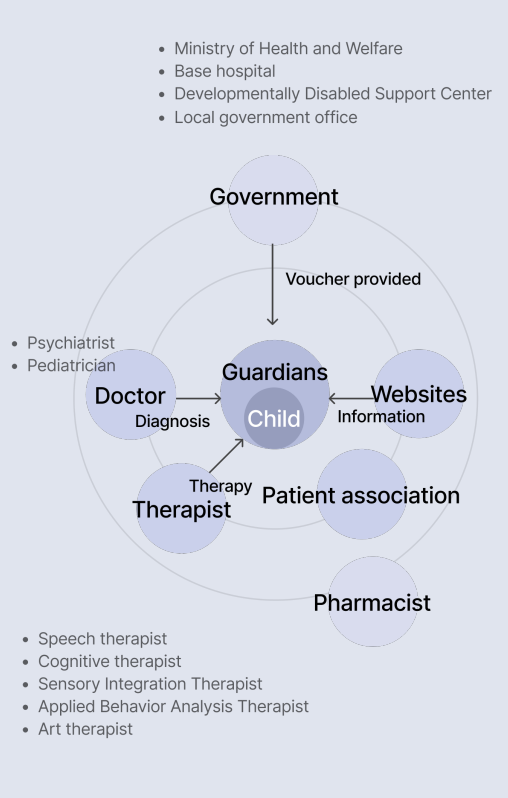
RESEARCH

Interviews revealed that parents were too busy, records were fragmented, and they didn't know what kind of data should be shared.

Desk Research : Surveys, Developmental Disability Blogs
Field Research : Qualitative research (interview - user/expert interview)
Data Modeling : Affinity Mapping, Personas
Service Concept : Brainstorming
Service Define : user scenario, storyboard, IA

Stakeholder Definition

Rather than treating caregivers as isolated users, we mapped the full ecosystem surrounding a child's care to understand where behavioral data breaks down across institutions, roles, and responsibilities

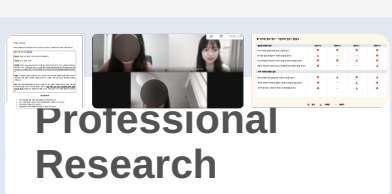

• Ministry of Health and Welfare
• Base hospital
• Developmentally Disabled Support Center
• Local government office
Government
Voucher provided
• Psychiatrist
• Pediatrician
Doctor
Diagnosis
Guardians
Child
Websites
Information
Therapy
Therapist
Patient association
Pharmacist
• Speech therapist
• Cognitive therapist
• Sensory Integration Therapist
• Applied Behavior Analysis Therapist
• Art therapist

In-Depth Interviews

To understand the needs of caregivers more deeply, we conducted 1:1 interviews with parents of children with ASD.

Participants: 6 parents of children with developmental disabilities Age of children: 3–5 (infancy) and 6–12 (childhood)

Method: One-on-one in-depth interviews, 35 minutes per session



Professional Research

helped us see caregiving as an ongoing negotiation between emotional labor and time pressure.

To understand the needs of caregivers more deeply, we conducted 1:1 interviews with parents of children with ASD.

Doctor Interviews

Participants: 2 child psychiatrists/ One-on-one interviews

Therapist Interviews

Participants: 4 developmental therapists (2 speech-language therapists, 2 occupational therapists)

Speaking with clinicians and therapists clarified a critical gap: while professionals value longitudinal behavioral patterns, the data they receive is often episodic, inconsistent, and shaped by caregiver stress rather than clinical relevance.

KEY INSIGHTS

Caregivers need an easier way to capture, share, and communicate a child's behavioral progress.

Guardians' parenting styles and emotional states affect their records

> "When a behavioral issue happens, I'm too busy handling the situation, removing dangerous objects or calming them down…"
> — Caregiver C

Multiple caregivers often share roles, leading to data fragmentation.

> "My child is sometimes cared for by their grandmother or a teacher, and in those cases, it's hard to manage all the records consistently."
> — Caregiver A

Uncertainty about what to report during short doctor visits

> "The consultation is so short, and I never know what I should show the doctor. I just freeze up."
> — Caregiver B

Data for both the doctor and the parents is needed. Track lack data for long-term treatment

> Experts emphasized the importance of offering the service in a digital format that enables doctors and therapists to easily review behavior logs and track progress.

Emotional Sensitivity of Caregivers

> Caregivers often experience significant emotional stress. The product should be designed to support their emotional well-being as well as provide practical tools.

FEATURE DEFINITIONS

Turning fragmented observations into shared, actionable records.

Guided logging for busy routines

Reduce documentation burden through quick, structured prompts that caregivers can complete during busy routines.

Shared behavior logs and schedule management

Enable multiple caregivers to collaboratively manage a shared schedule and behavioral records. This ensures all stakeholders are "on the same page" regarding the child's developmental status.

Visualized reports for clinical communication

Provide summarized, visual reports highlighting only the most relevant data for clinical use. This helps caregivers quickly and clearly communicate during short medical appointments.

UX WRITING

I defined writing principles to provide emotional warmth and reduce the burden of documentation.

Warm encouragement for emotionally burdened users

Parents of children with developmental disabilities experience the highest levels of stress among caregivers of children with illnesses or disabilities. I recognized that warm, encouraging communication is a crucial element in supporting consistent and sustainable record-keeping.

"You're already a wonderful mom, just by being present."

"Well done! You've finished all your logs for the week."

Ask questions like a kind, smart doctor

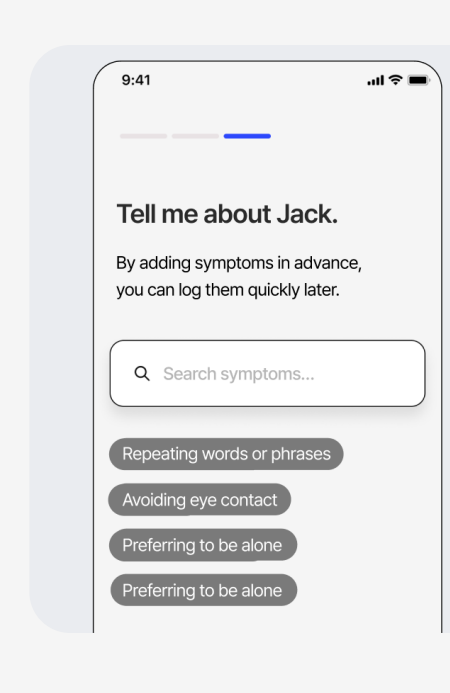
We designed our prompts to be simple and intentional—like a friendly doctor asking just the right questions—so that caregivers could feel guided, not overwhelmed.

"How did those responses from others make you feel?"

"Do these symptoms or behaviors occur repeatedly?"

SOLUTION 1

Quick Logs During Daytime, Detailed Updates Afterwards



9:41

Tell me about Jack.

By adding symptoms in advance, you can log them quickly later.

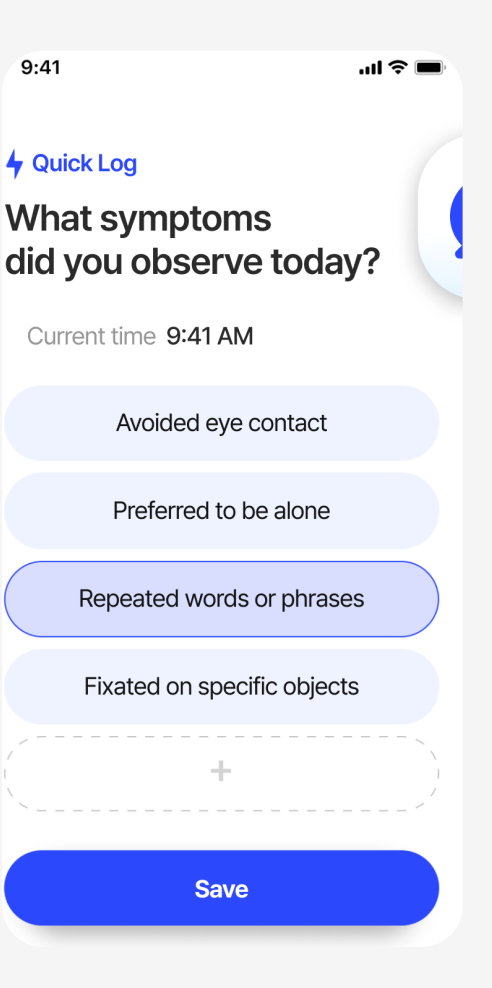
Search symptoms...
Repeating words or phrases
Avoiding eye contact
Preferring to be alone
Preferring to be alone

Symptom input during sign-up

Children with developmental disabilities often display a recurring set of symptoms. During sign-up, users can search for and select relevant symptoms that their child frequently exhibits. By categorizing these symptoms in advance, the system can later provide tailored content and personalized insights based on the selected symptom types.

Quick logging during daytime

When a symptom occurs, caregivers are often focused on managing the situation and unable to document it in detail. The Quick Log feature allows users to select a symptom with a single tap, automatically saving the selected symptom along with a timestamp—minimizing the burden of real-time data entry.



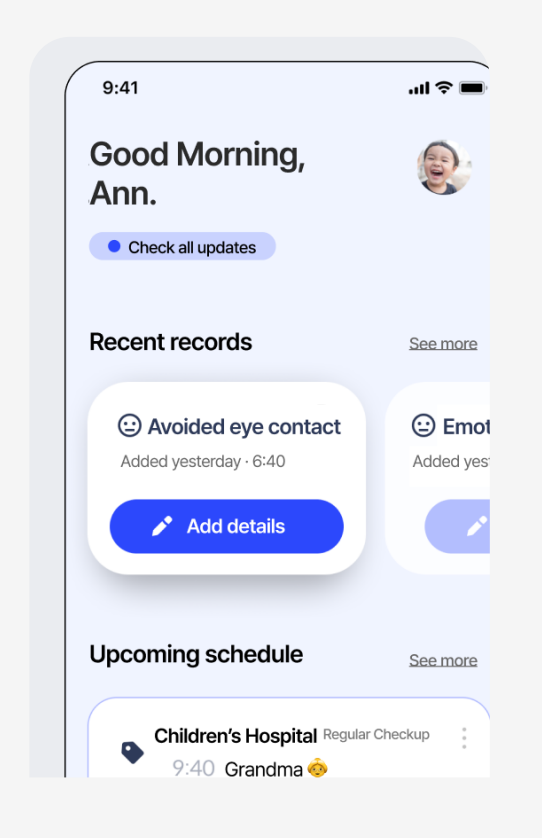
9:41

Quick Log

What symptoms did you observe today?

Current time 9:41 AM

Avoided eye contact
Preferred to be alone
Repeated words or phrases
Fixated on specific objects
+
Save



9:41

Good Morning, Ann.

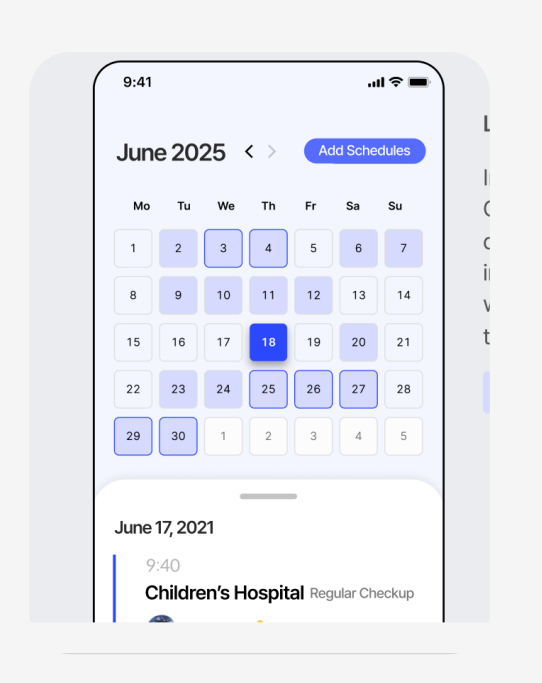
Check all updates
Recent records
See more
Avoided eye contact
Added yesterday · 6:40
Add details
Emot
Added yes
Upcoming schedule
See more
Children's Hospital Regular Checkup
9:40 Grandma

Detailed context logging

Records created via the Quick Log can be expanded later to include contextual information such as triggers, consequences, and calming strategies. This helps analyze behavioral patterns in more depth and creates a valuable record for treatment and professional consultation.

SOLUTION 2

Shared Behavior Logging Across All Caregivers



9:41

June 2025
Add Schedules
Mo Tu We Th Fr Sa Su
1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30 1 2 3 4 5
June 17, 2021
9:40
Children's Hospital Regular Checkup

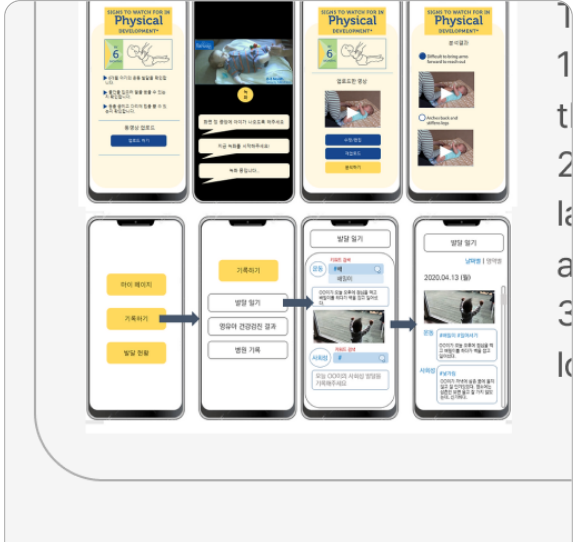
Logging Based on Shared Schedule

In most cases, parenting is not handled solely by the parents. Grandparents and childcare providers are often involved in the child's daily care. This leads to gaps in behavioral records and fragmented information. By linking behavior logging to a shared schedule, caregivers who spend time with the child are prompted to record observations in a timely and structured way.

- My schedule
- Others
- Today

User Testing with Target Users

A core-feature MVP was developed as an Android app and tested with target users through user testing (UT) sessions.


Physical
Physical
Physical

Test Scenarios

1) Log a problem behavior using the quick record feature
2) Add detailed notes and photos later as a developmental diary
3) Browse and review previously logged records

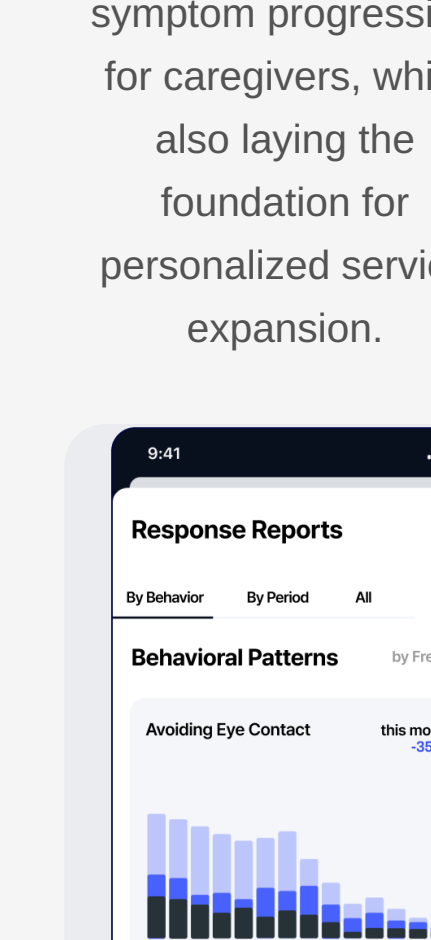
Step 2 (detailed input) was found to be quick and convenient. However, users found it difficult to locate and interpret past records intuitively.

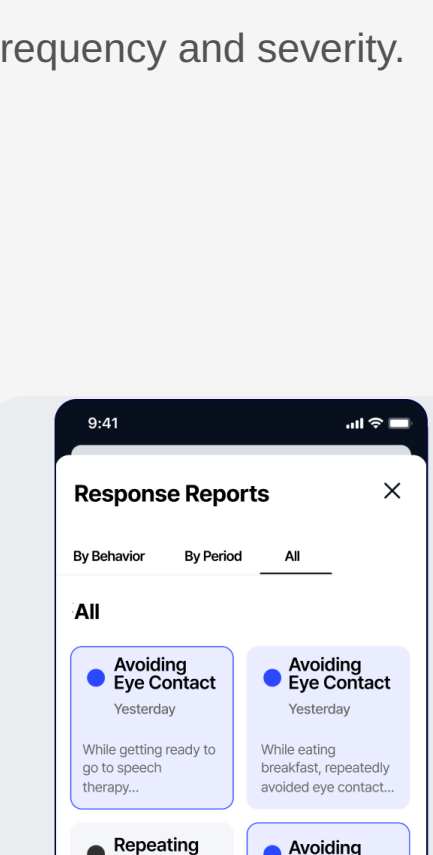
Building Objective Data

By allowing users to input the frequency of symptoms over defined time periods, the system accumulates objective data on behavioral patterns. This enables visual tracking of symptom progression for caregivers, while also laying the foundation for personalized service expansion.

Focused View for Clinical Use

The reporting feature prioritizes delivering only clinically relevant information in a short time frame. Informed by physician interviews, it allows filtering by symptom and time to surface trends in frequency and severity.


9:41
Response Reports
By Behavior
By Period
All
Behavioral Patterns
by Frequency
Avoiding Eye Contact
this month
-35 %
작년
5월
6월
Repeating Words
this month
-10 %


9:41
Response Reports
By Behavior
By Period
All
All
Avoiding Eye Contact
Yesterday
Avoiding Eye Contact
Yesterday
Repeating Words
Yesterday
Avoiding Eye Contact
Yesterday
Repeating Words
06.10
Repeating Words
06.10

While getting ready to go to speech therapy...

While eating breakfast, repeatedly avoided eye contact...

Went for a walk with the teacher...

Asked what they wanted to eat for breakfast...

Choosing shoes by the shoe cabinet...

Opened the closet to change clothes...